ADEM YAHYA

Embedded Systems and IoT Engineer

+216 58 78 58 89 | ademyahya01@gmail.com | LinkedIn

PROFILE

Dynamic and passionate Embedded Systems Engineer with a strong analytical mindset and excellent adaptability. Rigorous, autonomous, and organized, I work effectively in teams to achieve set objectives.

EDUCATION

2023 – 2026 Engineering Degree in Embedded Systems and IoT — ISI Manar, Tunis
2020 – 2023 Bachelor's Degree in Embedded Systems and IoT — ISSAT Gafsa
2020 Technical Baccalaureate — Ahmed Snnousi High School, Gafsa

PROFESSIONAL EXPERIENCE

Final Year Project — Embedded Vision for ADAS Systems *Feb. 2026 – Jun. 2026*
Capgemini Engineering, Tunisia

- Developed and optimized YOLOv8 object detection models using Knowledge Distillation.
- Accelerated performance on FPGA (ZedBoard) with VHDL / Vitis HLS implementation.
- Optimized models in reduced precision (FP16) and deployed on Jetson Nano.
- Developed a visualization and control interface in Python (PyQt / Qt).
- Technologies: PyTorch, NumPy, OpenCV, Ultralytics YOLO, FPGA, VHDL, Jetson Nano.

Internship — Smart Car Key Security *Jun. 2025 – Aug. 2025*
Capgemini Engineering, Tunisia

- Developed a secure car key system resistant to RF attacks (Replay, Relay, Injection).
- Implemented on Raspberry Pi 4, STM32F4, and Arduino with 433 MHz RF modules.
- Developed a Flutter mobile application integrating security algorithms (AES, RSA, DES).

Introductory Internship *Jun. 2024*
Streamlink, Tunis

- Developed a desktop application to control a robot via STM32 (UART).
- Developed in Python (PyQt) with serial communication.

Bachelor's Final Year Internship *Feb. 2023 – May 2023*
Orange Tunisia, Tunis

- Designed and modernized the NB-IoT network architecture.
- Testing and validation using Atoll software.

ACADEMIC PROJECTS

- **CAN/Ethernet Gateway for Connected Car:** Designed a remote climate-control system (STM32F7 Discovery, Raspberry Pi 4, CAN, Ethernet, MQTT Broker, Flutter).
- **Facial Recognition Access Control on Embedded Linux (Yocto):** Developed a custom Linux image with Yocto for Raspberry Pi 4, real-time face detection and recognition (Python, TensorFlow, OpenCV).
- **Driver Drowsiness Detection System:** Mobile monitoring of driver alertness (Python, OpenCV, Raspberry Pi, Flutter).
- **16-bit RISC Processor:** Design and implementation (VHDL, Datapath, Control Unit).
- **Autonomous Mobile Robot:** Obstacle detection and avoidance with mapping (SLAM), ROS2/Gazebo simulation with A*, deployed on Jetson Nano.

SKILLS

Programming C, C++, Python, C#, VHDL, Assembly
Embedded Systems STM32, ESP32, Raspberry Pi, Arduino, Jetson, FPGA, FreeRTOS, ROS2
AI / Vision YOLOv8, TensorFlow, PyTorch, OpenCV, Knowledge Distillation, FP16, A*, LLM, RAG
Development Flutter, Qt, PySide6, HTML, CSS, JavaScript, PHP, SQL
DevOps / Tools Linux, Git, GitHub, Docker, Jenkins, Ansible, SAP ERP
Methodologies Agile Scrum, PDCA, Jira

CERTIFICATIONS

- Azure AI Fundamentals
- Microsoft Excel
- Cisco IoT
- Cisco DevNet
- Network Security (VPN & Firewall)

LANGUAGES

- Arabic: Native
- French: Advanced
- English: Advanced

EXTRACURRICULAR ACTIVITIES

- **Enactus —** Financial Vice-President (2022 – 2026).
- **Freeways —** Robotics Club Member (2023 – 2026).